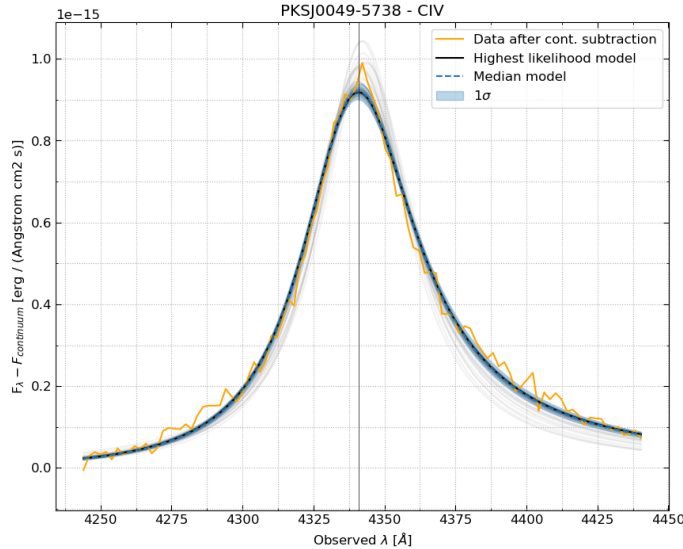


easyspec News

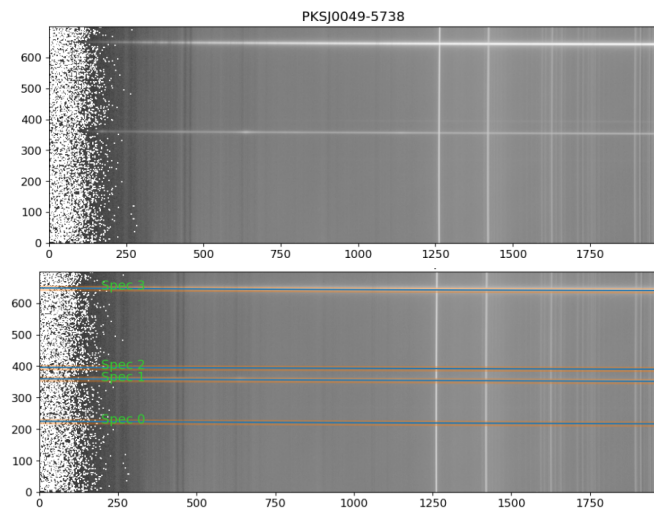


Skewed line profiles

Check out the new **skewed Lorentzian** and **skewed Gaussian** models now available in *easyspec*. In this implementation, the parameterization is expressed in terms of the **line peak**, rather than the traditional Lorentzian or Gaussian mean. Detailed descriptions of these parameterizations are available in the documentation, and a new tutorial demonstrates the wide range of applications enabled by this feature.

Tracing continuum-free spectra

Some astrophysical objects, such as **HII regions**, are often observed without an underlying continuum. In such cases, a common technique consists of aligning the slit with both the HII region and a nearby point-like source, then repeating the trace of the point-like source at the expected y -position of the target within the slit.



In the figure above, *easyspec* identifies three spectra using the command `extraction.tracing(target_data,`

`method="multi", peak_height=120, ..., repeat_trace_at={"spec_0":-135})`, and then repeats the bottom trace 135 pixels below using the dictionary `repeat_trace_at={"spec_0":-135}`. The user can then extract the spectrum from this new region, even when no continuum is present.

Major speed boost in the MCMC

The Lorentzian, Gaussian, and Voigt models are now significantly faster in the MCMC, thanks to a vectorization of the code. Keep in mind, however, that skewed models still require considerably more time to run.

Build your synthetic spectrum

In the new tutorial, we show how to build your own synthetic spectrum based on MCMC fitting. In the figure below, we show the synthetic lines CIV + He II + O III].

